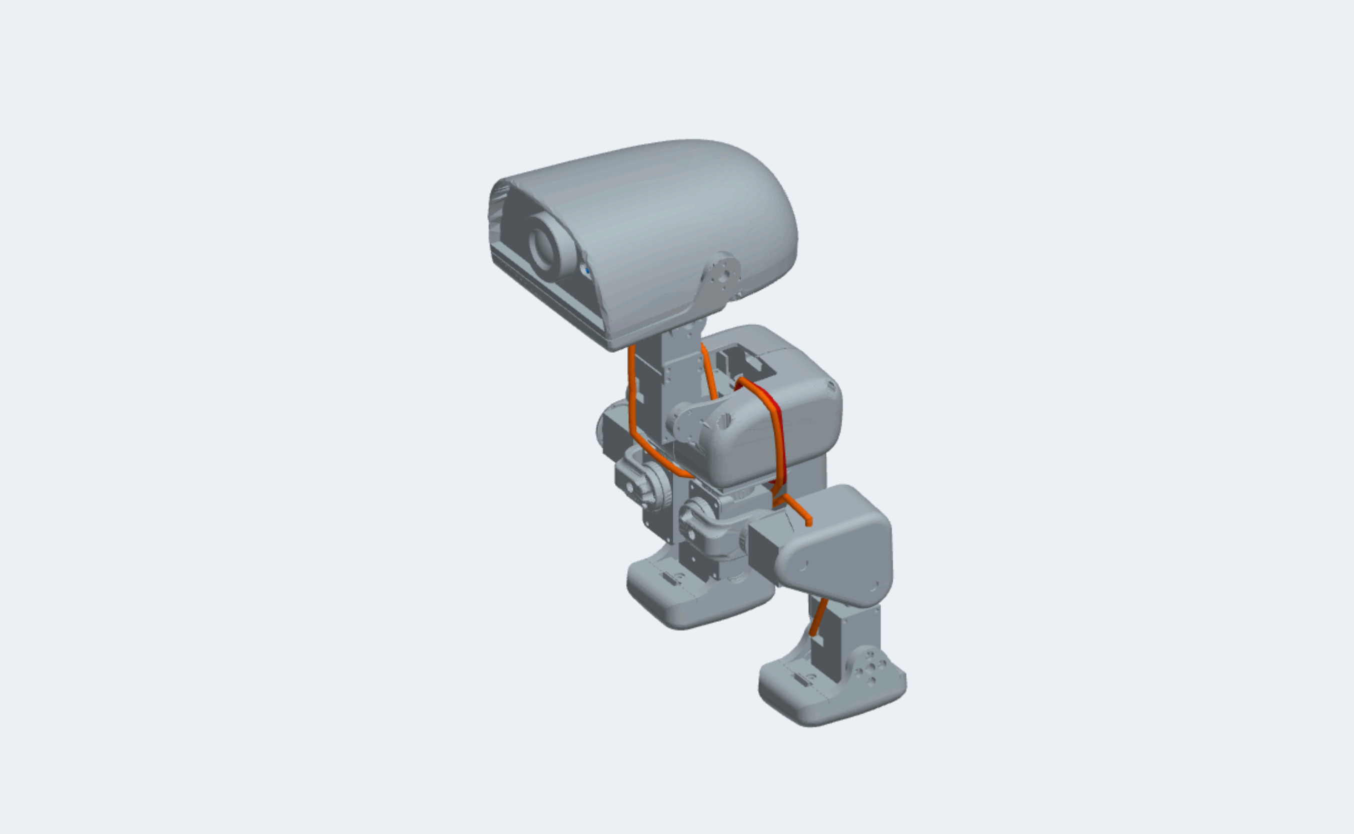
Microduck

Cables & connections / 线缆与连接

Review the wiring inside our current robot. The pictures use the actual published robot meshes and 16 saved cable models. Click a picture to rotate it offline in Edge or Chrome.

查看当前机器人内部接线。图片采用实际已发布机器人网格和16根候选线缆模型。点击图片可在浏览器中离线旋转。

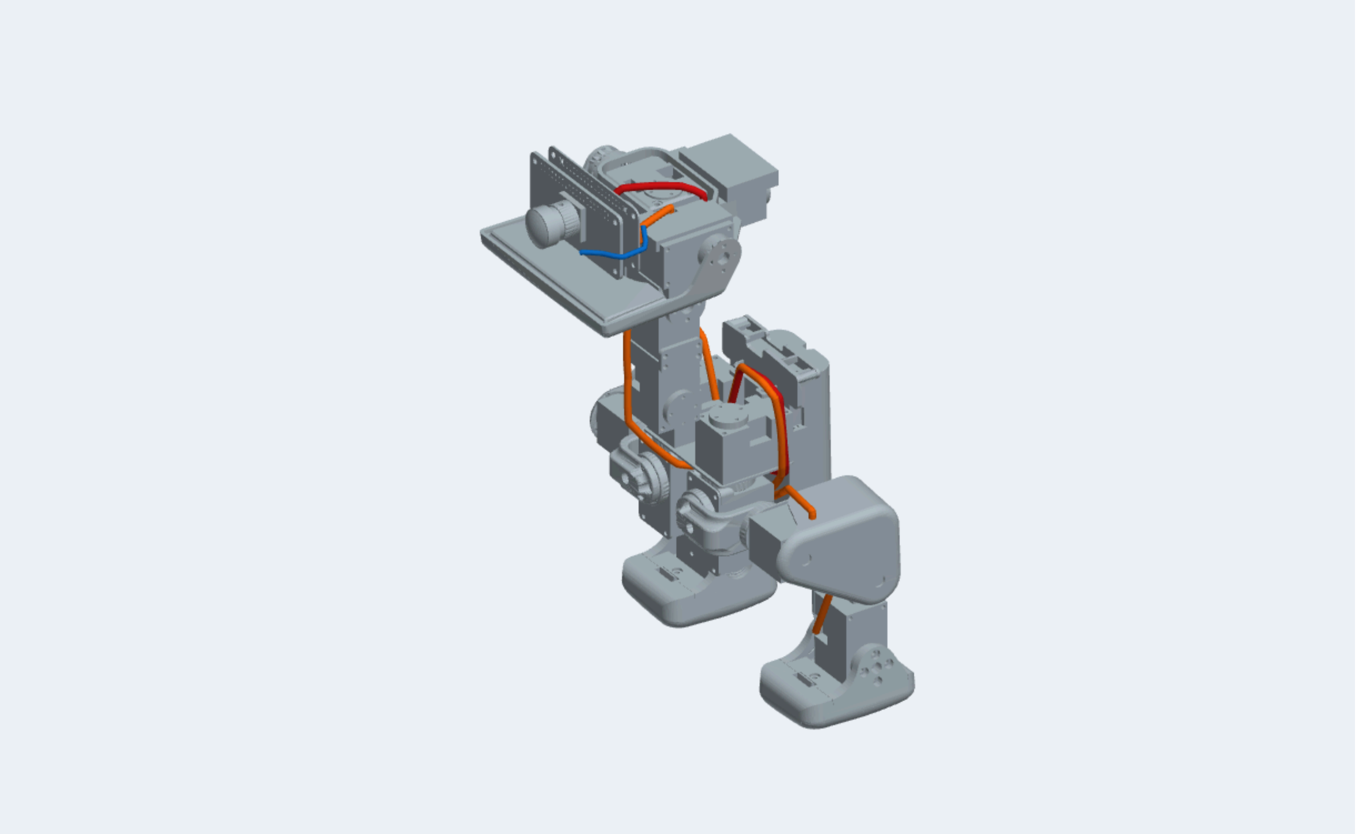
> **Review stage — do not power the robot from this guide yet.**
> **审查阶段：本指南暂不能用于通电操作。**
>
> The current design still uses 15 ROBOTIS Dynamixel XL330-M288-T servos. No cheap-servo substitution has been approved. The public HAT connects servo power to battery voltage; XL330 is rated 3.7–6.0 V. The modeled 2S supply exceeds that rating. The power stage and actual board revision must be resolved first. The public HAT audit also records 40 DRC errors and 9 warnings. [PCB quotation file map / PCB报价文件核对](#).
>
> 当前仍使用15个原规格XL330舵机，尚未采用廉价替代。公开HAT舵机供电连接电池，而XL330额定电压为3.7–6.0 V；当前2S电源模型超出额定值，必须先解决供电与板卡版本问题；公开HAT检查还记录40项DRC错误和9项警告。



Grey: real robot geometry. Orange: unverified candidate routes. Red: routes with a recorded failure. Blue: distance sensor lead. 灰色为机器人；橙色为待验证线缆，红色为已有失败记录，蓝色为测距传感器线。 [Turn this view / 旋转查看]

1. Open the covers / 先打开外壳

Identify the battery in the body, the computer and HAT in the head, and the numbered servo chain. Keep the battery disconnected while checking connector identity and alignment. 图中显示电池、头部计算机与HAT，以及舵机链。确认接头和方向时保持电池断开。



Covers removed from actual geometry. Cable diameter is a historical nominal model value, not a measured jacket diameter. 隐藏实际外壳网格；线径为候选值，尚未经实物测量。 Turn this view / 旋转查看

2. Follow the servo chain / 核对舵机链

Head / 头部

HAT → ID34 → ID33 → ID32 → ID31 → ID30 → IMU ID200

Left leg / 左腿

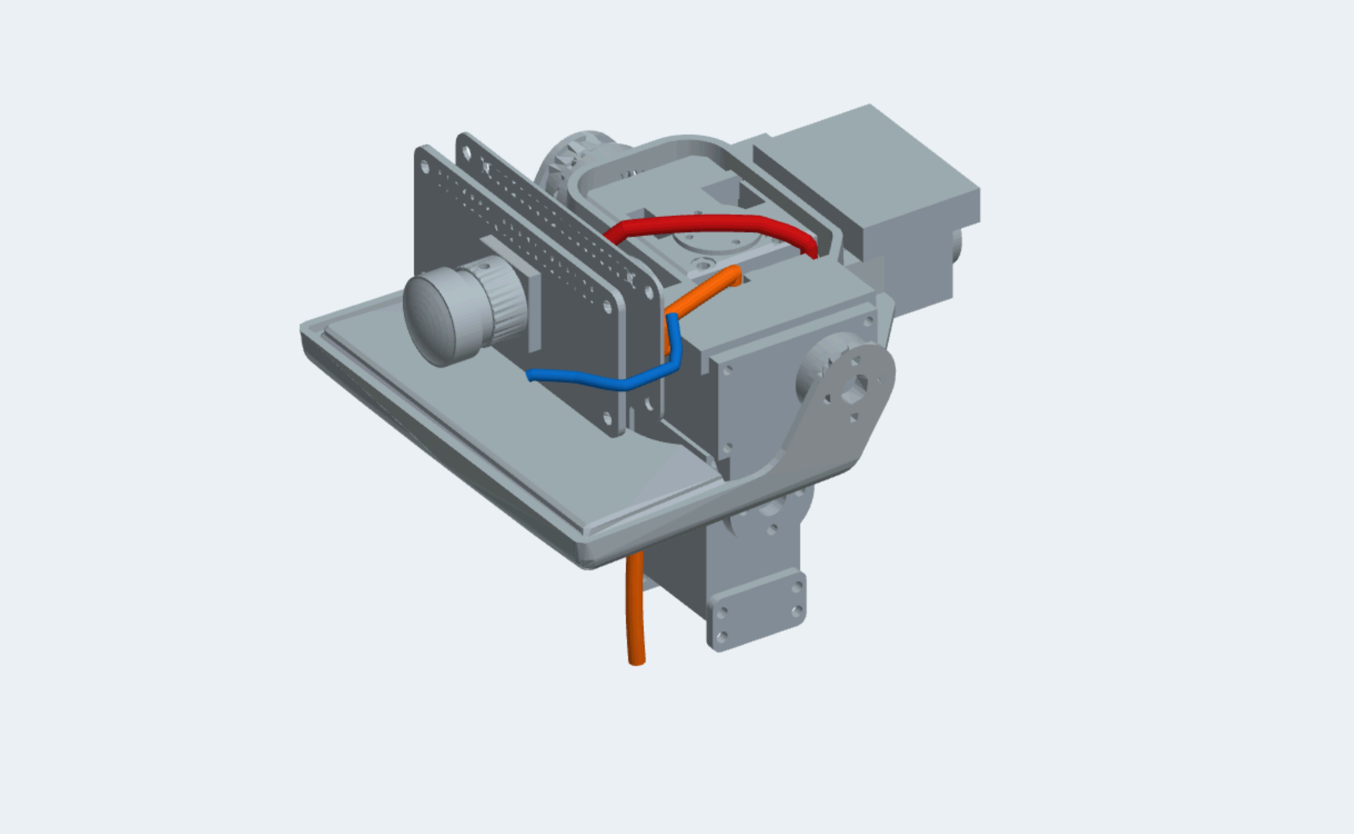
IMU ID200 → ID20 → ID21 → ID22 → ID23 → ID24

Right leg / 右腿

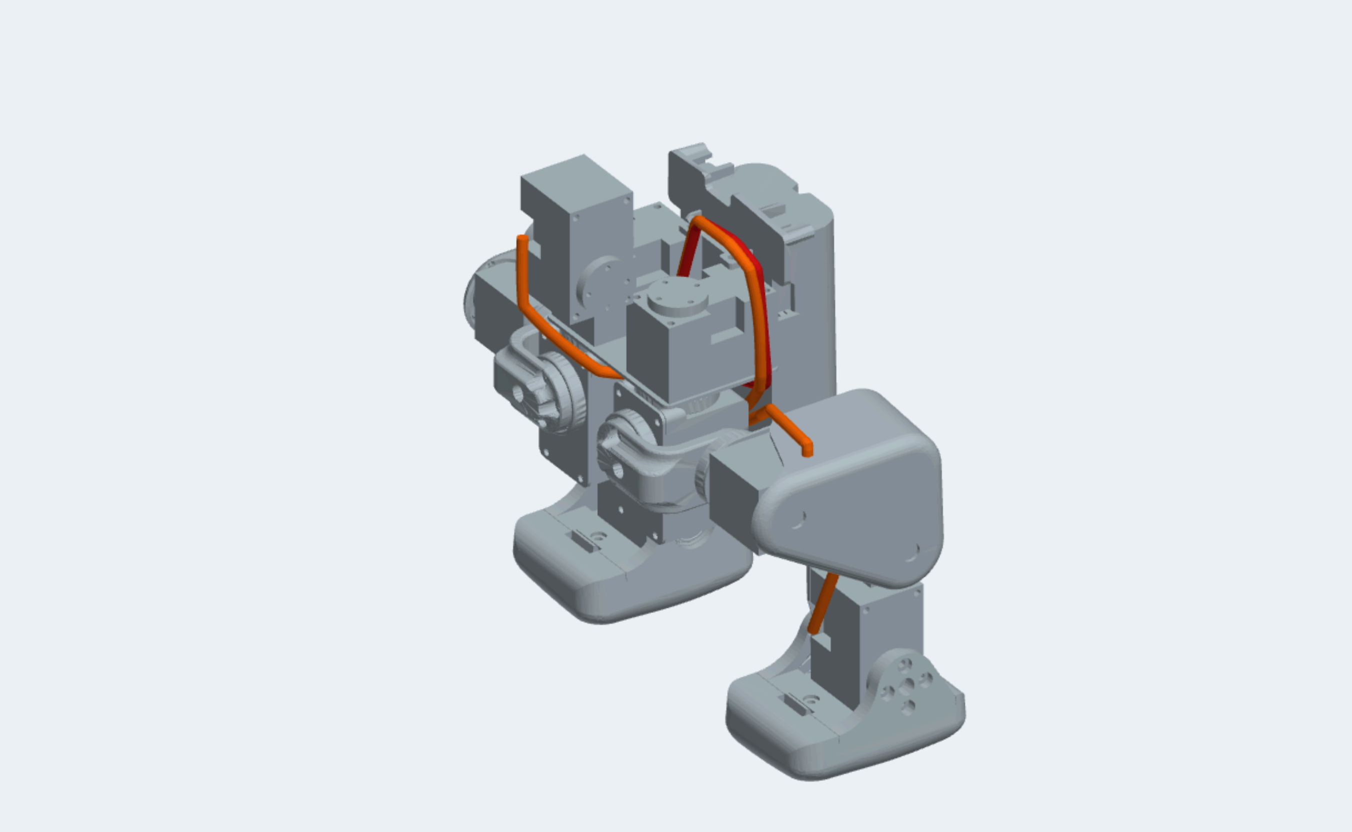
IMU ID200 → ID10 → ID11 → ID12 → ID13 → ID14

This is the existing proposed cable topology, not a proven factory harness. The IMU junction board and its socket allocation remain unconfirmed. The ID33–ID32 connection has no successful saved route. 上述为现有候选连接拓扑，并非已验证原厂线束；IMU接头分配未确认，ID33至ID32尚无成功路由。

XL330 cable pin numbers: 1 GND · 2 VDD · 3 DATA. JST EH 3-pin, EHR-03 housing. Use connector pin numbers; a plug viewed from its wire side is mirrored relative to its mating side. This page does not assign a left-to-right orientation from an unknown photograph. 舵机线：1地、2电源、3数据；按接头编号检查，不凭照片左右顺序判断。



Head and neck: inspect the routes at the moving joints. The cable model does not prove slack or moving clearance. 头颈运动关节需进一步验证余量与动态间隙。 Turn this view / 旋转查看



Legs: inspect the visible bends before closing the covers. No cable cut length is approved. 双腿弯折处需要检查，目前没有批准的裁线长度。 Turn this view / 旋转查看

3. Connect the head electronics / 核对头部电子连接

Connection / 连接	Public source says / 公开资料	Still to verify / 仍需确认
HAT ↔ Radxa Zero 3W	2×20 header; power pins 2/4 = 5 V. UART pins 8/10; I²C pins 3/5. Board-to-board connection, no cable.	Match pin 1 and all 40 pins with the actual board drawing. Do not rotate the board solely because screw holes align. 对照实板确认1号针及40针位置。
ToF distance sensor / 测距传感器	Public HAT J5: pin 1 GND, 2 +3V3, 3 SDA, 4 SCL; JST SH 4-pin.	Exact sensor model, sensor-end connector and cable orientation unknown. 传感器型号、另一端接头及方向未确认。
Speaker / 扬声器	Public HAT J1 is the differential speaker output via amplifier/filter, Wago terminal footprint.	Original speaker impedance, termination and polarity labeling need confirmation. Neither speaker terminal is a ground return. 阻抗与端接未确认，两个输出端均不应当作地线。

Camera / 摄像头	Radxa side uses a 22-pin 0.5 mm CSI ribbon connector.	Camera identity, opposite-end pin count, contacts-facing direction and adapter unknown. No ribbon polarity guessed. 摄像头端针数、触点方向及转接线未确认。
Microphone / 麦克风	Public HAT already has an onboard microphone; J2/J9 are optional external paths.	External microphone is not automatically required. Confirm actual board first. 先确认实物板，不能默认需要外置麦克风。
Battery / 电池	NP-F550 contact arrangement through a contact board is modeled.	Contact hardware, polarity, protection and servo regulator are unresolved. Do not wire raw 2S directly to XL330 under this guide. 电池触点、保护与舵机稳压尚未解决。

4. Before closing or powering / 合壳和通电前

1. Confirm the exact HAT, IMU, camera and ToF board revisions against their drawings. 确认各板卡实际版本。
2. Verify each connector pin number, voltage and polarity with the matching source drawing; label both cable ends. 按对应图纸核对针脚、电压、极性，并标记线缆两端。
3. Resolve the servo voltage mismatch before connecting the battery. 先解决舵机供电超压问题。
4. With power disconnected, check slack and pinching through the intended joint range; retain loops and strain relief only after clearance is measured. 断电检查全行程余量、夹线和固定位置。
5. Measure the actual routed lengths, then finalize cable cuts. The current pictures do not provide cut lengths. 实际走线测量后才能确定裁线长度。

These are the 16 saved candidate cable solids, with no new routes invented for missing wires. 本图为16根已保存候选线缆，没有虚构缺失线缆。 Turn this view / 旋转查看

Every connection / 全部连接

All 23 records are shown, including one descriptive port row with quantity zero and the direct board header. Download the connection list / 连接清单.

Cable	From → To	Saved route	Physical length
dxl-hat-id34	hat → id34	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id34-id33	id34 → id33	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id33-id32	id33 → id32	No routed cable model / 无路由模型 FAIL	Not established / 未确定
dxl-id32-id31	id32 → id31	Rendered candidate / 已显示候选 FAIL	Not established / 未确定

dxl-id31-id30	id31 → id30	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id30-imu200	id30 → imu200	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-imu200-id20	imu200 → id20	Rendered candidate / 已显示候选 FAIL	Not established / 未确定
dxl-id20-id21	id20 → id21	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id21-id22	id21 → id22	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id22-id23	id22 → id23	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id23-id24	id23 → id24	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-imu200-id10	imu200 → id10	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id10-id11	id10 → id11	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id11-id12	id11 → id12	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id12-id13	id12 → id13	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
dxl-id13-id14	id13 → id14	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
tof-hat	hat → tof	Rendered candidate / 已显示候选 CANNOT DETERMINE	Not established / 未确定
spk-hat	hat → speaker	No routed cable model / 无路由模型 FAIL	Not established / 未确定
mic-hat	hat → mic	No routed cable model / 无路由模型 CANNOT DETERMINE	Not established / 未确定

csi-radxa-camera	radxa → camera	No routed cable model / 无路由模型 FAIL	Not established / 未确定
bat-hat	battery → hat	No routed cable model / 无路由模型 FAIL	Not established / 未确定
hat-radxa-40pin	hat → radxa	No routed cable model / 无路由模型 CANNOT DETERMINE	Not established / 未确定
hat-dxl-port	hat → id34	No routed cable model / 无路由模型 CANNOT DETERMINE	Not established / 未确定

Where this comes from / 来源与版本

Our robot geometry and saved cable routes are recorded by SHA-256 in PROVENANCE.json. The public HAT is pinned to revision `23eab11927f95ceca0dfa35bf182caeb7db39ea0`; pin evidence is read back from its KiCad PCB. The HAT mesh and released board differ: a historical J14 projection differs by 49.75 mm. The images retain this discrepancy for review rather than presenting it as an approved connection.

fanhao375/microduck-replica at 2549257 documents printed structural parts and screws fitting the leg parts. Its README explicitly says the remaining work is analysis; this is not evidence of a complete working replica. That repository is Git-versioned. Our review records are separately preserved, with exact source hashes. The original robot's software, public board designs and simulation meshes have different release/licensing scopes.

参考项目公开记录了结构件打印和腿部螺钉试装，不等于整机已完成验证。上游采用Git版本管理，本次审查另保存来源哈希。实物板版本、柔性线缆运动及生产就绪程度仍未验证。